Thermodynamics and Statistical Physics

7th Exercise Class

Micro- and macrostates; Boltzmann entropy, Gibbs theorem and Gibbs paradox; Microcanonical ensemble

Exercises for the Class

1. A system consists of six distinguishable particles, each of which can occupy three energy levels:

$$E_0 = 0, \quad E_1 = \epsilon, \quad E_2 = 2\epsilon.$$

   (a) Determine the number of microstates for the macrostates corresponding to $E = 2\epsilon$ and $E = 4\epsilon$.

   (b) Assuming all microstates are equally probable, compute the probability of these macrostates.

2. (Gibbs theorem) A semipermeable membrane is one that allows the passage of one type of molecule. At equilibrium, the partial pressures by the gas on either side of such a membrane are equal. Such membranes exist.

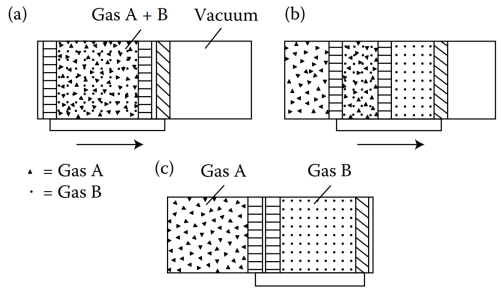


Fig. 1

   Consider a mixture of two ideal gases A and B contained in the left-hand half of the box, as shown in Fig. 1 (a). There is a vacuum in the right-hand half. The box is fitted with a pair of coupled sliding pistons; the left-hand one is permeable to A only, while the right-hand one is impermeable to both. The box is also divided into two halves with a semipermeable membrane, permeable to B only.

   Now slide the coupled pistons slowly to the right as shown in Fig. 1 (b) so that, eventually, the two gases separate reversibly. They will finally each occupy a volume equal to the original volume of the mixture. This is shown in Fig. 1 (c). Let this process occur isothermally.

   (a) Show that the net force on the coupled pistons is zero.

   (b) The heat flowing into the system in this isothermal reversible process is given by:

$$Q = T(S_f - S_i)$$

   where S_i and S_f are the initial and final entropies. By applying the first law of thermodynamics, show that:

$$S_i = S_f$$

   This implies the Gibbs theorem, which states that the entropy of a gas of two components is the sum of the entropies of the components at the same temperature occupying the volume alone:

$$S_{A+B}(T, V) = S_A(T, V) + S_B(T, V)$$

   (c) Prove the Gibbs theorem for the Boltzmann entropy.

3. (Gibbs paradox)

   Assume that n_A moles of an ideal gas A of volume V_A and temperature T are separated from n_B moles of another ideal gas B of volume V_B at the same temperature T. The partition is removed so that the gases mix isothermally at the temperature T, and the mixture then occupies the volume $V_A + V_B$.

   (a) Use the Gibbs theorem, introduced in the preceding problem to calculate the entropy change due to the mixing.

   (b) Suppose that the gases are identical. Is the result of (a) valid for identical gases, and if not, why not and what is the correct formula for the entropy change of mixing?

4. Consider a microcanonical system of N noninteracting, distinguishable particles, where each particle can be either in its ground state with zero energy, or in a twofold degenerate excited state with energy ϵ.

   (a) Calculate the number of microstates $\Omega(E, N)$ corresponding to a microcanonical macrostates.

   (b) Express the entropy $S(E, N)$.

   (c) Derive expressions for temperature from the entropy. Discuss the result when

$$E \geq \frac{2N\epsilon}{3}.$$

(d) Calculate the specific heat of the system.

(e) Show that the change of the thermodynamic entropy $\delta Q_{\mathrm{R}}/T$ is the same as the change of the statistical entropy.

Homework to Be Submitted

1. Consider a system of N distinguishable particles in a microcanonical ensemble at energy E, where the number of accessible microstates $\Omega(E)$ is given by:

$$\Omega(E) = \left(\frac{E}{\epsilon}\right)^{N/2} \exp\left(-\frac{E}{E_0}\right)$$

where ϵ is the energy of a single microstate, and E_0 is a characteristic energy scale.

(a) Calculate the Boltzmann entropy $S(E)$.

(b) Determine the temperature $T(E)$ of the system.

(c) Calculate the heat capacity C_V of the system. Discuss the behavior of the heat capacity as $T \to 0$ and $T \to \infty$.

Homework for Practice

1. A system of four weakly interacting distinct particles is such that each particle can be in one of four states with energies $\epsilon, 2\epsilon, 3\epsilon$ and 4ϵ respectively. If the system has total energy 15ϵ, what is the entropy of the system? For what possible values of total energy is the entropy of the system zero?

2. Let us consider the first law of thermodynamics for a system with volume V, number of particles N, energy E, and entropy $S = S(E, V, N)$. Using the definition of the chemical potential as

$$\mu = \left(\frac{\partial F}{\partial N}\right)_{T,V}$$

with the free energy given by $F = E - TS$, and the relation

$$\left(\frac{\partial S}{\partial E}\right)_{N,V} = \frac{1}{T}$$

prove that

$$\frac{\mu}{T} = -\left(\frac{\partial S}{\partial N}\right)_{E,V}$$

To solve the problem, use the differential expression of $S = S(E, V, N)$ with the volume V and the temperature T kept constant.

3. A physical system is composed of N distinguishable particles, and each particle can be found in a state with energy 0 or $\varepsilon > 0$. The excited state has degeneracy $d = 4$ while the ground state is not degenerate. The total energy of the system is given by $E = n\epsilon$, with n a positive integer ($n \leq N$). Write down the number of microstates corresponding to the macrostate with energy $E = n\varepsilon$. Then, identify the temperature T and compute the ratio of the occupation numbers for the two energy levels as a function of T and ε. Verify the limit of high temperatures in the final result.